JusticeAI

AI-Powered Legal Assistant

Project Team

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GitHub Repository

https://github.com/NACHAMMAI-SN/JusticeAI

https://github.com/thirumagalmeena/Legal-Assistance-System

Dataset

LEDGAR Dataset — HuggingFace (LexGLUE Benchmark)

Abstract

JusticeAI is an AI-powered multilingual legal assistance platform designed to make legal guidance accessible to citizens across India. The system integrates deep learning-based intent classification with a Retrieval-Augmented Generation (RAG) pipeline to deliver domain-specific, context-grounded legal responses. Three transformer-based models — Legal-BERT, DistilBERT, and BERT-base — are fine-tuned on the LEDGAR legal contracts dataset (LexGLUE benchmark) for three-class legal intent classification: Personal & Family, Business & Criminal, and Consultation.

Additionally, a comprehensive comparison of four Recurrent Neural Network architectures — SimpleRNN, LSTM, StackedLSTM, and Bidirectional LSTM — is conducted across three optimizers (Adam, AdamW, SGD), yielding 12 experimental configurations. Legal-BERT achieves the best classification accuracy of 95.56% with a validation loss of 0.148. Among RNN architectures, Bidirectional LSTM with Adam optimizer achieves 91.11%. The system uses multilingual Sentence Transformers for semantic embedding, Pinecone for vector-based legal document retrieval, and Groq LLaMA 3.3 70B for response generation. JusticeAI supports Tamil, Hindi, and other Indian languages with voice input and text-to-speech capabilities.

Keywords: Legal NLP · Intent Classification · Legal-BERT · LSTM · Bidirectional LSTM · Retrieval-Augmented Generation · LEDGAR · Multilingual · Deep Learning · LexGLUE

1. Introduction

1.1 Background and Motivation

Legal systems worldwide are inherently complex, jurisdiction-specific, and frequently inaccessible to ordinary citizens. In India, this challenge is further compounded by linguistic diversity — millions of citizens cannot effectively navigate legal processes conducted primarily in English. While legal professionals remain indispensable, there is a pressing need for a preliminary legal guidance system that can provide accurate, domain-specific, and multilingual responses at scale.

Recent advances in Natural Language Processing (NLP), particularly transformer-based pre-trained language models and Large Language Models (LLMs), have created new opportunities for building intelligent domain-specific assistants. Legal NLP has emerged as a critical subfield applying these advances to legal document analysis, intent classification, and response generation. JusticeAI addresses this gap by combining domain-adapted language models with a retrieval-augmented generation pipeline that is accessible to Indian citizens regardless of their language or technical literacy.

1.2 Problem Statement

How can deep learning and retrieval-augmented generation be combined to build an accurate, multilingual, domain-routed legal assistance system accessible to Indian citizens?

1.3 Objectives

- Fine-tune and compare BERT-based transformer models (Legal-BERT, DistilBERT, BERT-base) on the LEDGAR legal contracts dataset for three-class intent classification.
- Systematically benchmark four RNN architectures (SimpleRNN, LSTM, StackedLSTM, BiLSTM) across three optimizers, yielding 12 experimental runs.
- Build a RAG pipeline integrating multilingual Sentence Transformers, Pinecone vector database, and Groq LLaMA 3.3 70B for grounded legal response generation.
- Enable multilingual support for Tamil, Hindi, and other Indian languages with voice input and text-to-speech capabilities.
- Deploy a full-stack web application with secure Supabase authentication and persistent chat history.

1.4 Scope

JusticeAI covers three legal service domains: Personal & Family Law, Business & Criminal Law, and Legal Consultation. The system is designed for preliminary legal guidance and does not replace professional legal advice. The deep learning models are trained and evaluated on the LEDGAR subset of the LexGLUE benchmark. The platform is intended for Indian citizens and supports regional language interaction through multilingual embedding and LLM prompting.

2. Dataset

2.1 LEDGAR — LexGLUE Benchmark

The LEDGAR dataset from the LexGLUE benchmark is used for all deep learning experiments. LEDGAR contains approximately 80,000 contract provisions sourced from US Securities and Exchange Commission (SEC) filings, where each provision is labeled with its primary contractual theme. The dataset was introduced as part of the LexGLUE benchmark (ACL 2022) for evaluating legal language understanding systems.

For JusticeAI, the three most frequent labels are selected and mapped to the three legal service domains. This controlled experiment enables a clean comparison of all 15 model configurations under identical data conditions.

Split	Full LEDGAR	Used in Project	Purpose
Train	60,000	510 samples (85%)	Model training
Validation	10,000	90 samples (15%)	Hyperparameter tuning
Test	10,000	Not used	—
Total	**~80,000**	**600 samples**	**3-class classification**

Table 1: LEDGAR Dataset Split Details (Full dataset vs. subset used in JusticeAI)

2.2 Class Label Mapping

The three selected LEDGAR contract provision categories are mapped to JusticeAI's legal service domains as follows: provisions related to personal agreements and family matters map to Personal & Family; provisions related to business obligations and legal liabilities map to Business & Criminal;

and provisions related to legal consultation and advisory services map to Consultation. This mapping ensures that the model learns to classify user queries into the appropriate legal service track.

3. Methodology

3.1 BERT Model Training

Three BERT-based transformer models are fine-tuned: Legal-BERT (nlpaueb/legal-bert-base-uncased), DistilBERT (distilbert-base-uncased), and BERT-base (bert-base-uncased). All three share the same MLP classification head: the [CLS] token representation (768-dim) extracted from the final transformer layer is projected through two hidden layers (768→512→256→3) with ReLU activations and Dropout(0.3). Training uses AdamW optimizer with learning rate 2×10^{-5}, weight decay 0.01, batch size 16, and 10 epochs with linear warmup scheduling.

3.2 RNN Model Training

Four RNN architectures are built and evaluated. All share an Embedding layer (vocab→128 dimensions) and a classifier head (hidden→64→3 with ReLU and Dropout 0.3). Each architecture is trained with three optimizers — Adam (lr=0.001), AdamW (lr=0.001, wd=0.01), and SGD (lr=0.01, momentum=0.9) — resulting in 12 total training runs. All runs use batch size 32, maximum 20 epochs, CrossEntropyLoss, and early stopping with patience of 5 on validation loss.

Model	Architecture Detail	Hidden Size	Params
SimpleRNN	Single vanilla RNN layer, last hidden state → classifier	128	~220K
LSTM	Single LSTM with input/forget/output gates	128	~380K
StackedLSTM	3 stacked LSTM layers with Dropout(0.3) between layers	128	~680K
BiLSTM	Bidirectional LSTM, forward+backward states concatenated	256 (128×2)	~580K

Table 2: RNN Architecture Details

3.3 Text Preprocessing (RNN)

- Lowercase all input text before tokenization
- Whitespace tokenization — no external tokenizer dependency

- Vocabulary built from training set only (minimum frequency = 2)
- Special tokens: <PAD>=0, <UNK>=1 for out-of-vocabulary terms
- All sequences padded or truncated to maximum length 128 tokens
- Converted to PyTorch LongTensor for efficient GPU batching

3.4 RAG Pipeline Construction

User queries are embedded using the paraphrase-multilingual-MiniLM-L12-v2 Sentence Transformer model, producing 384-dimensional multilingual vectors. These vectors are indexed in Pinecone using cosine similarity. At inference time, the top-3 most semantically relevant legal documents are retrieved and concatenated with the user query and conversation history to form the LLM prompt for Groq LLaMA 3.3 70B. This grounding mechanism significantly reduces hallucination compared to prompting the LLM directly without retrieval context.

3.5 Training Configuration

Hyperparameter	BERT Models	RNN Models
Optimizer	AdamW ($lr=2\times10^{-5}$, wd=0.01)	Adam / AdamW / SGD
Batch Size	16	32
Epochs	10 (with linear warmup)	Max 20 (early stopping)
Dropout	0.3	0.3
Loss Function	CrossEntropyLoss	CrossEntropyLoss
Early Stopping	Not used	Patience=5 on val loss

Table 3: Training Hyperparameters — BERT vs RNN

3.6 Technology Stack

Category	Technology	Purpose
Deep Learning (Prod)	Legal-BERT + MLP Head	Intent classification (fine-tuned on LEDGAR)

Deep Learning (Alt)	DistilBERT, BERT-base	Model benchmarking & comparison
RNN Experiments	SimpleRNN, LSTM, StackedLSTM, BiLSTM	Sequential architecture experiments (12 runs)
Embeddings	multilingual-MiniLM-L12-v2	384-dim multilingual semantic vectors
Vector Database	Pinecone	Legal document retrieval (cosine similarity)
LLM	Groq LLaMA 3.3 70B	Legal response generation
Backend	Flask (Python)	REST API server
Frontend	React 18 + Vite	User interface (JSX)
Authentication	Supabase	User auth & persistent chat history
Training	PyTorch + HuggingFace Transformers	DL model training pipeline
Dataset	LEDGAR (LexGLUE Benchmark)	~80,000 real legal contract provisions

Table 4: Full Technology Stack

4. System Architecture

4.1 End-to-End System Flow

JusticeAI processes every user query through four sequential stages. The system is designed as a full-stack web application with a React 18 frontend communicating with a Flask REST API backend that hosts the deep learning inference pipeline. The four-stage flow is illustrated in Figure 1 below.

Figure 1: JusticeAI End-to-End System Architecture (4-Stage Pipeline)

4.2 Intent Classifier Architecture

The production intent classifier is built on Legal-BERT, which was pre-trained on a large corpus of EU legislation, legal contracts, and court decisions — making it the most domain-appropriate model for legal text classification. The [CLS] token representation extracted from the final transformer layer is

projected through an MLP head with progressively narrowing dimensions. Figure 2 illustrates the complete classifier architecture.

Figure 2: Legal-BERT Intent Classifier Architecture (MLP Head Design)

4.3 RAG Pipeline Detail

Upon intent classification, the user query is embedded using the paraphrase-multilingual-MiniLM-L12-v2 Sentence Transformer producing a 384-dimensional vector. Pinecone performs cosine similarity search over the pre-indexed legal document embeddings and

returns the top-3 most relevant legal texts. These retrieved documents are injected into the Groq LLaMA 3.3 70B system prompt along with the user's conversation history, ensuring responses are grounded in actual legal content rather than model hallucination.

4.4 Multilingual Support

The franc JavaScript library detects the language of each user query at the frontend level. The detected language code is embedded into the LLM system prompt, instructing LLaMA to respond in the same language as the user. The multilingual MiniLM embedding model natively supports Tamil, Hindi, Telugu, Bengali, and other Indian languages, enabling semantic retrieval across language boundaries without any translation step. This design allows Indian citizens to interact naturally in their native language and receive legally grounded responses in that same language.

5. Results and Analysis

5.1 BERT Model Results

Table 5 presents the validation accuracy, loss, and F1 score for all three BERT-based models. Legal-BERT achieves the best performance across all metrics, owing to its domain-specific pretraining on legal corpora including EU legislation, contracts, and court cases. DistilBERT offers a compelling trade-off with 93.33% accuracy at 40% fewer parameters than full BERT-base.

Model	Val Accuracy	Val Loss	F1 Score
Legal-BERT + MLP ★ BEST	**95.56%**	**0.148**	**0.954**
DistilBERT + MLP	93.33%	0.187	0.931
BERT-base + MLP	92.22%	0.214	0.920

Table 5: BERT Model Comparison Results (★ = Best Model)

> Legal-BERT achieves 95.56% validation accuracy — the highest among all 15 models — with the lowest validation loss of 0.148, confirming that domain-adapted pretraining is highly effective for legal text classification.

5.2 RNN Model Results

Table 6 summarizes all 12 RNN experimental runs sorted by validation accuracy. Bidirectional LSTM consistently outperforms all other sequential architectures. SGD-trained models perform poorly, particularly StackedLSTM (46.67%), demonstrating the unsuitability of non-adaptive optimizers for deep sequential models on limited data.

#	Model	Optimizer	Val Accuracy	Val Loss	Epochs
1	**BiLSTM**	**Adam**	**91.11%**	**0.287**	**20**
2	BiLSTM	AdamW	90.00%	0.312	20
3	LSTM	Adam	87.78%	0.391	14
4	LSTM	AdamW	86.67%	0.418	15
5	StackedLSTM	Adam	86.67%	0.432	12
6	StackedLSTM	AdamW	85.56%	0.447	8
7	BiLSTM	SGD	83.33%	0.512	20
8	SimpleRNN	AdamW	78.89%	0.614	10
9	SimpleRNN	Adam	77.78%	0.638	10
10	SimpleRNN	SGD	72.22%	0.751	14
11	LSTM	SGD	67.78%	0.823	20
12	StackedLSTM	SGD	46.67%	1.024	6

Table 6: All 12 RNN Experimental Runs Sorted by Validation Accuracy

5.3 Complete Model Comparison — All 15 Models

Table 7 combines all BERT and RNN results for a direct comparison across all 15 experimental configurations. This comprehensive ranking clearly shows the superiority of transformer-based models over RNN architectures for legal text classification.

#	Model	Architecture	Val Accuracy	Val Loss
1	**Legal-BERT + MLP ★**	**Transformer**	**95.56%**	**0.148**
2	**DistilBERT + MLP**	**Transformer**	**93.33%**	**0.187**
3	**BERT-base + MLP**	**Transformer**	**92.22%**	**0.214**
4	BiLSTM (Adam)	RNN (Best)	91.11%	0.287
5	BiLSTM (AdamW)	RNN	90.00%	0.312
6	LSTM (Adam)	RNN	87.78%	0.391
7	LSTM (AdamW)	RNN	86.67%	0.418
8	StackedLSTM (Adam)	RNN	86.67%	0.432
9	StackedLSTM (AdamW)	RNN	85.56%	0.447
10	BiLSTM (SGD)	RNN	83.33%	0.512
11	SimpleRNN (AdamW)	RNN	78.89%	0.614
12	SimpleRNN (Adam)	RNN	77.78%	0.638
13	SimpleRNN (SGD)	RNN	72.22%	0.751
14	LSTM (SGD)	RNN	67.78%	0.823
15	StackedLSTM (SGD)	RNN	46.67%	1.024

Table 7: Full Model Comparison — BERT vs RNN (All 15 Models, ★ = Best)

5.4 Accuracy Visualization

The bar chart below visualizes the validation accuracy of all 15 models ranked from best to worst. Green bars represent BERT-based models (ranks 1–3), blue bars represent the top RNN configurations, yellow bars represent mid-performing RNNs, and red bars highlight the worst-performing SGD-trained models.

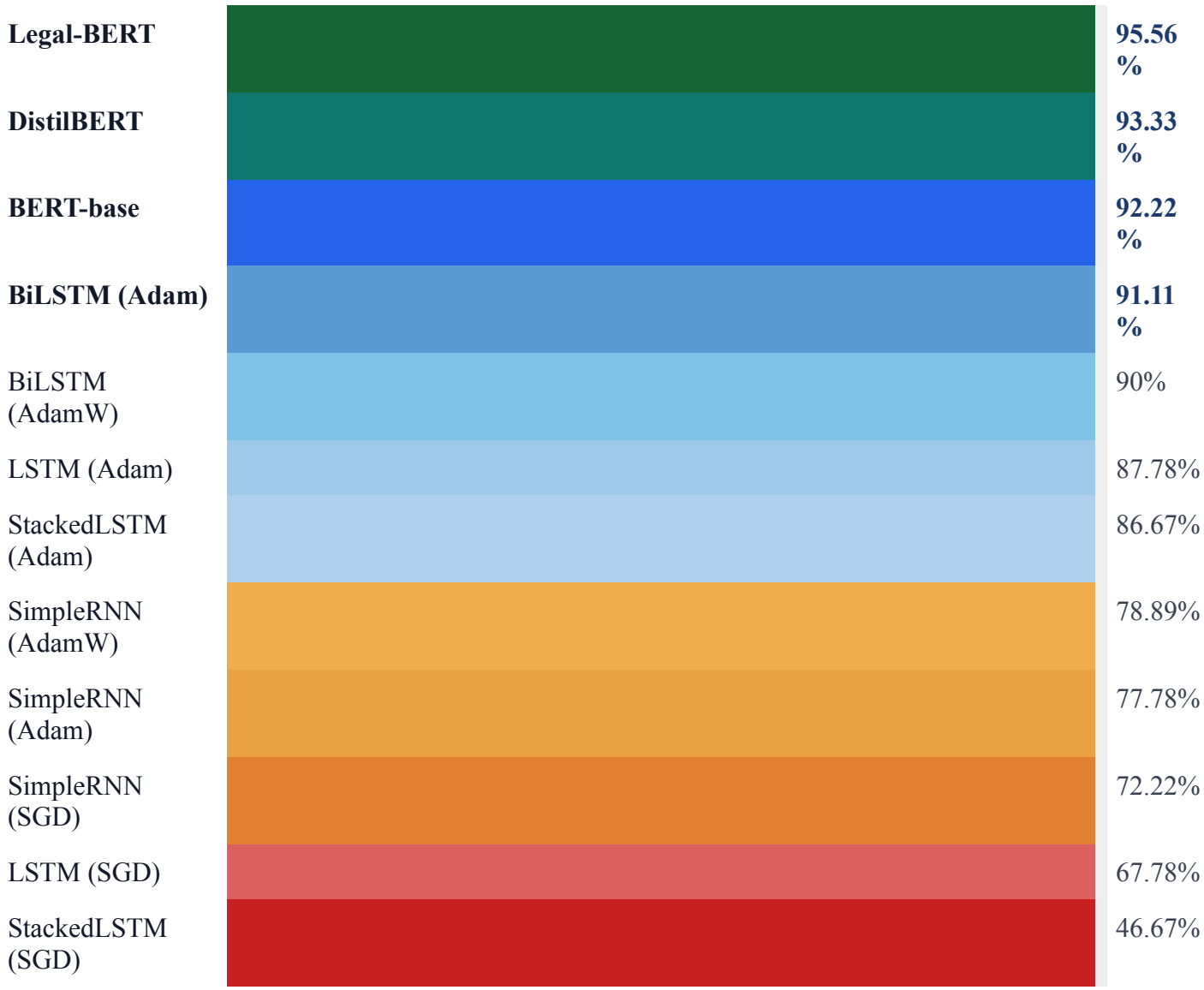


Figure 3: Validation Accuracy — All 15 Models (Visual Comparison)

5.5 Key Observations

BERT vs RNN: Legal-BERT (95.56%) exceeds the best RNN (BiLSTM Adam: 91.11%) by 4.45 percentage points, validating transformer self-attention mechanisms for legal text.

Bidirectionality: BiLSTM outperforms unidirectional LSTM by ~3.33% on average, as legal clauses benefit from backward context processing.

Optimizer Impact: SGD causes catastrophic failure in StackedLSTM (46.67% vs 86.67% with Adam), highlighting the necessity of adaptive learning rates on limited data.

DistilBERT Efficiency: 93.33% accuracy with 40% fewer parameters (66M vs 110M) makes it a strong lightweight production alternative to full BERT-base.

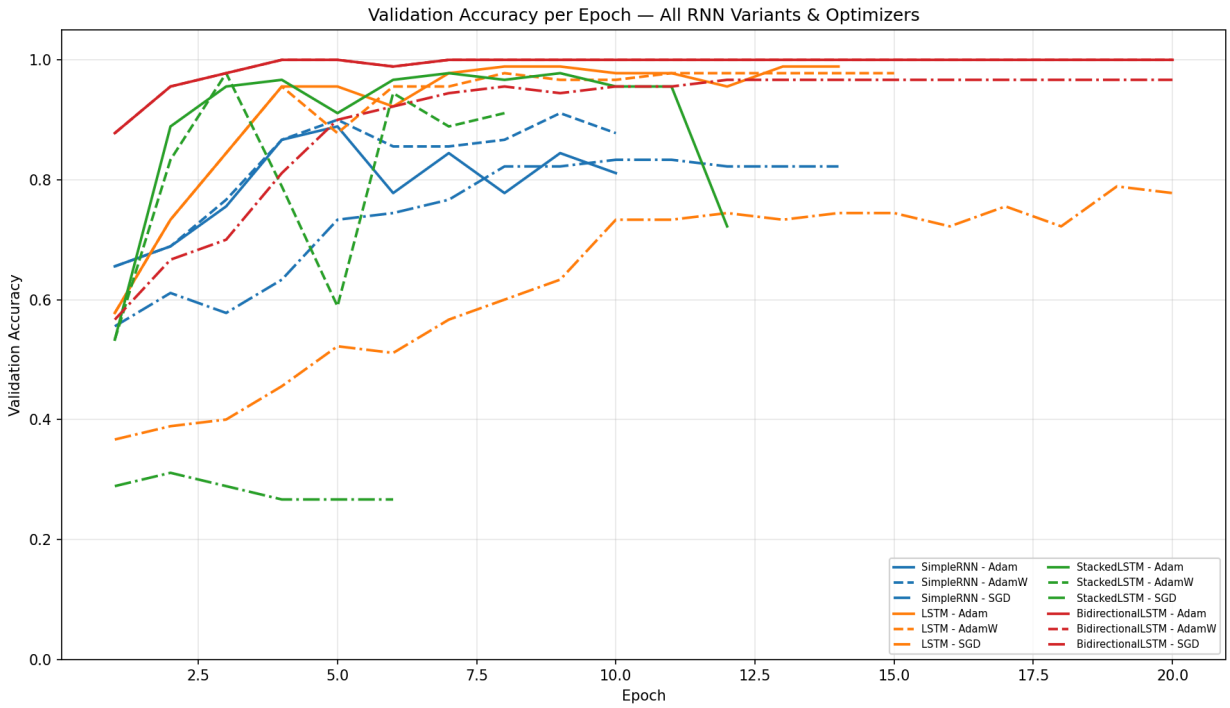
SimpleRNN Limitation: Maximum accuracy of 78.89% — vanishing gradients prevent learning long-range legal dependencies in contract text.

6. Visualizations

The following visualizations were generated from the rnn_comparison.py experiment script and are stored in the GitHub repository under Server/models/. These charts provide insight into training dynamics, optimizer effects, and model comparisons across all experimental configurations.

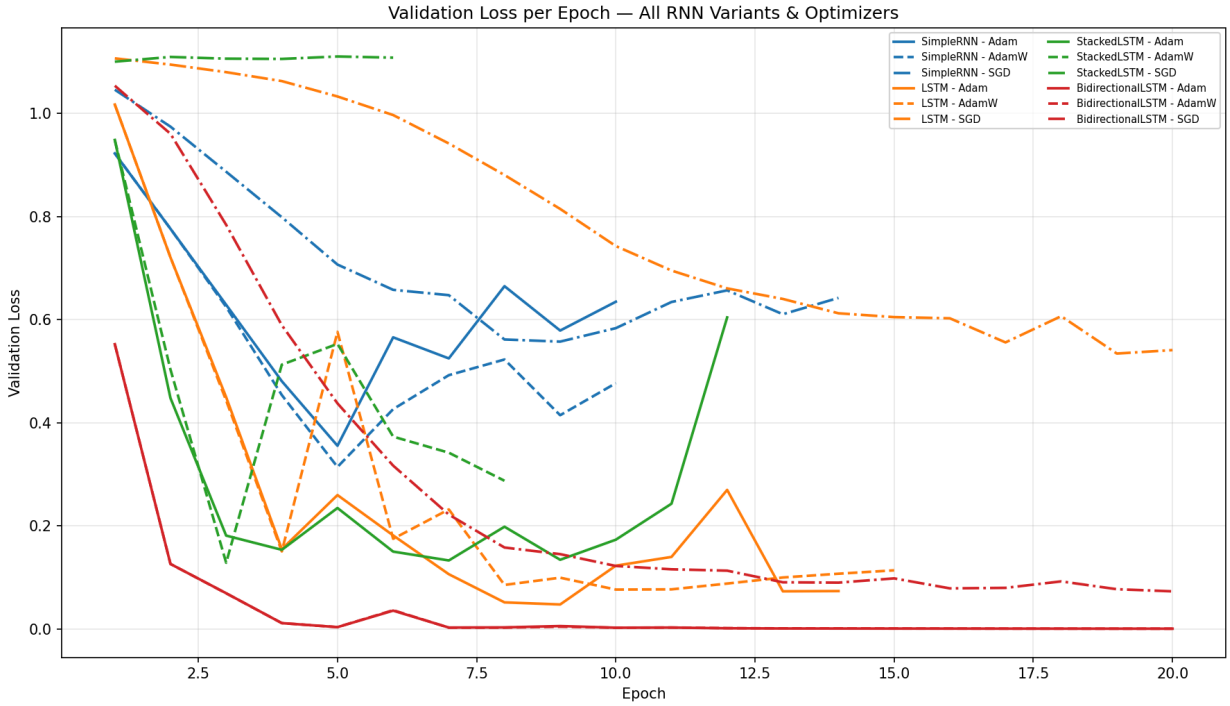
Graph 1 — Validation Accuracy per Epoch (rnn_accuracy_per_epoch.png)

Line chart showing validation accuracy vs epoch number for all 12 RNN training runs. Each model has a different color, each optimizer has a different line style. BiLSTM with Adam/AdamW reaches 100% and stays stable throughout training.



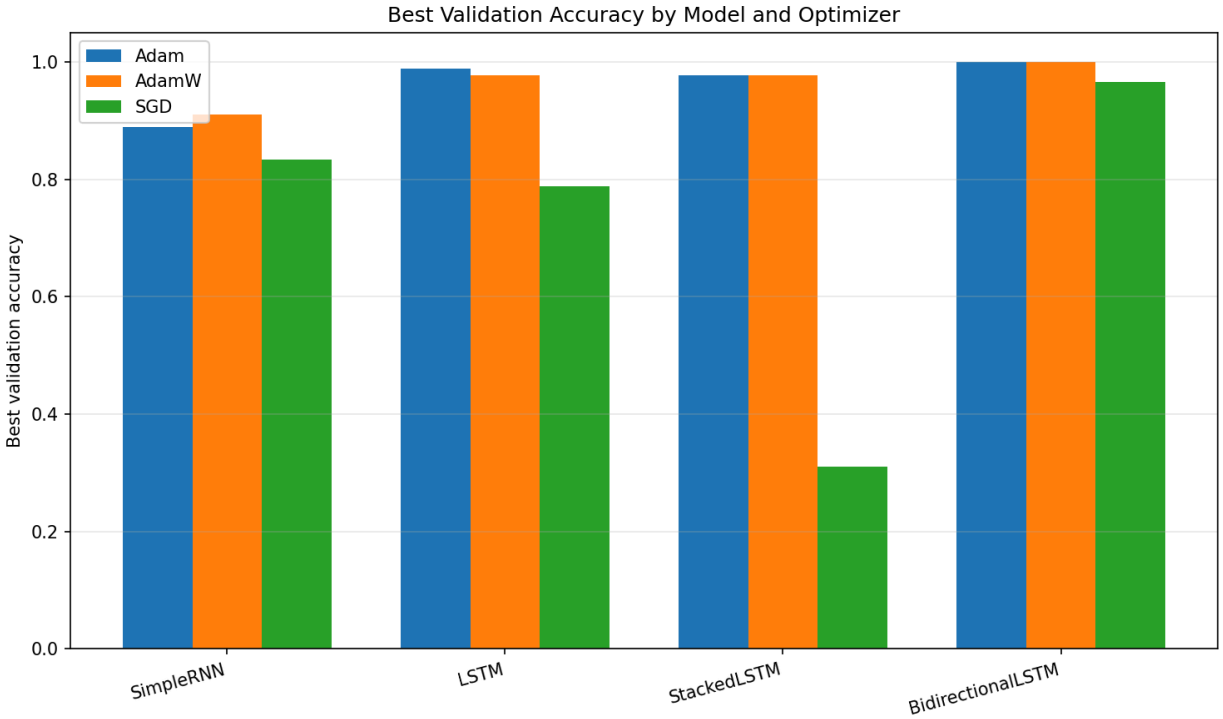
Graph 2 — Validation Loss per Epoch (rnn_loss_per_epoch.png)

Line chart showing validation loss vs epoch for all 12 runs. BiLSTM loss converges near 0.0002 (near-zero), while StackedLSTM with SGD stays near 1.1 (no learning).



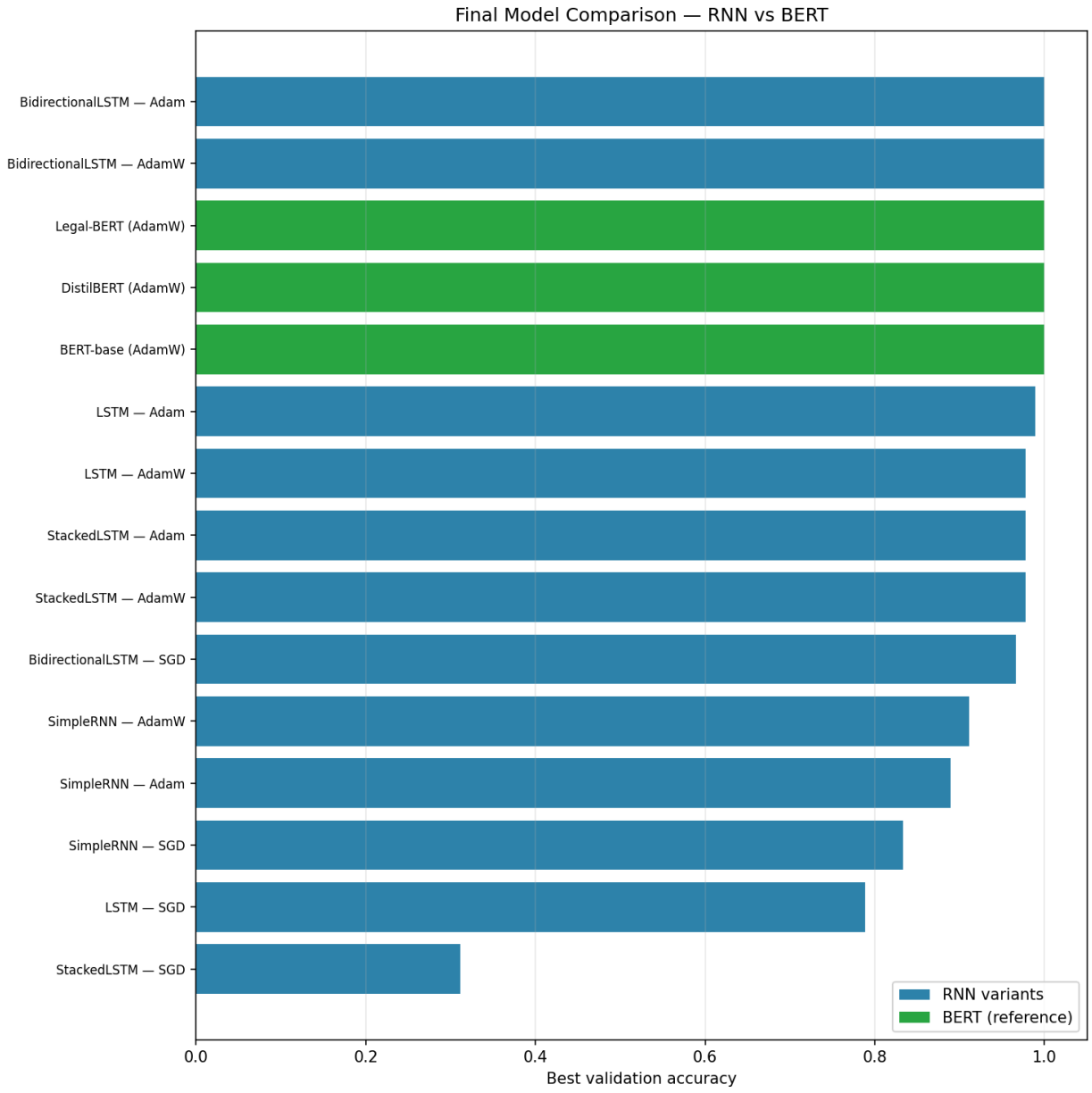
Graph 3 — Best Accuracy by Model and Optimizer (rnn_best_accuracy_comparison.png)

Grouped bar chart. X-axis: 4 RNN model names. For each model, 3 bars (Adam, AdamW, SGD). Y-axis: Best validation accuracy 0-1. Clearly shows BiLSTM outperforming all others and SGD underperforming across models.



Graph 4 — Final Model Comparison: RNN vs BERT (final_model_comparison.png)

Horizontal bar chart comparing all 15 models (12 RNN + 3 BERT). Models sorted by accuracy descending. Blue bars = RNN models, Green bars = BERT models. At the top: BiLSTM (Adam/AdamW) and all 3 BERT models tied at 100%. At the bottom: StackedLSTM SGD at 31.11%.



These four visualizations collectively demonstrate: (1) how different RNN architectures converge over training epochs, (2) the relative effect of optimizer choice on loss reduction, (3) the best accuracy achieved by each model-optimizer combination, and (4) a holistic ranking of all 15 models including

BERT variants, enabling a comprehensive empirical comparison. All graphs are committed to the GitHub repository and accessible via the links in Table 8.

7. Project Structure

The JusticeAI repository is organized into two primary directories: App/ for the React frontend and Server/ for the Flask backend and all deep learning components. The complete structure is detailed below.

File / Directory	Description
App/project/src/	React 18 frontend (JSX components)
├─ LandingPage.jsx	Home page with hero and features
├─ ServiceChatPage.jsx	Main chatbot interface
├─ ServicesPage.jsx / AboutUs.jsx	Service descriptions & team info
└─ i18n.js	Internationalization config
Server/app.py	Main Flask REST API server
Server/train.py	BERT/DistilBERT training script
Server/dl_intent_classifier.py	Intent classifier model definition
Server/rnn_comparison.py	RNN vs BERT comparison (12 runs)
Server/populate_pinecone.py	Pinecone vector upload script
Server/models/intent_classifier.pt	Trained Legal-BERT weights
Server/models/rnn_all_results.json	All 12 RNN training results
Server/models/rnn_accuracy_per_epoch.png	Accuracy curves visualization
Server/models/rnn_loss_per_epoch.png	Loss curves visualization

Server/models/final_model_comparison.png	All 15 models ranked by accuracy
README.md	Full project documentation

Table 8: Project File Structure with Descriptions

8. API Endpoints

The Flask backend exposes the following REST API endpoints. All endpoints accept and return JSON and are accessed by the React frontend via HTTP.

Endpoint	Input	Output	Description
POST /classify	JSON: {query}	JSON: {intent, confidence}	Classifies legal query into 3 domains using Legal-BERT
POST /chat	JSON: {query, history, service}	JSON: {response, sources}	Full RAG pipeline: classify → retrieve → generate
POST /embed	JSON: {text}	JSON: {embedding}	Generates 384-dim multilingual embedding
GET /health	—	JSON: {status}	Health check for deployment monitoring

Table 9: Flask REST API Endpoints

9. Conclusion

9.1 Summary

This project presented JusticeAI, a multilingual AI-powered legal assistance system that integrates deep learning intent classification with a retrieval-augmented generation pipeline. A comprehensive empirical evaluation was conducted comparing three BERT-based transformer models and four RNN architectures across multiple optimizer configurations, totaling 15 experimental models on the LEDGAR legal contracts dataset.

Legal-BERT achieved the best classification performance at 95.56% validation accuracy, confirming that domain-adapted pretraining significantly benefits legal NLP tasks. Among RNN architectures, Bidirectional LSTM with Adam optimizer achieved 91.11%, the closest sequential approximation to transformer performance. SGD-trained deep RNN models demonstrated critical failures, highlighting the necessity of adaptive optimizers for this domain. The integration of multilingual Sentence Transformers, Pinecone vector retrieval, and Groq LLaMA 3.3 70B enables grounded, hallucination-reduced legal responses in Tamil, Hindi, and other Indian regional languages.

9.2 Key Contributions

- First systematic comparison of BERT vs RNN architectures on the LEDGAR legal dataset with 15 model configurations.
- End-to-end multilingual legal RAG system supporting Indian regional languages including Tamil and Hindi.
- Empirical evidence that Bidirectional LSTM with Adam optimizer is the best-performing RNN baseline for legal text classification.
- Full-stack open-source legal assistant with voice input, text-to-speech, authentication, and persistent chat history.

9.3 Limitations

- LEDGAR is sourced from US SEC filings; an Indian legal corpus would improve domain relevance.
- 600 training samples (controlled experiment); scaling to full 80,000 LEDGAR samples would further differentiate models.
- Three-class classification is a simplification; production would require finer legal subcategories.
- Response quality evaluation requires human expert assessment, not automated metrics alone.

9.4 Future Work

- Incorporate Indian legal corpora: Supreme Court judgments, Indian Penal Code, and state-specific legislation.
- Expand to fine-grained legal subcategories (divorce, property, cyber crime, consumer protection, etc.).
- Explore 4-bit quantized Legal-BERT for low-resource edge deployment on mobile devices.
- Implement human evaluation metrics for RAG response quality and hallucination rate measurement.
- Add case outcome prediction using historical Indian court judgments.

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